

Banking System Using Abstract Data Types (ADT)

ADT

- Abstract Data Type
- Defines WHAT operations can be done

Operations

- Deposit
- Withdraw
- Balance Inquiry
- Transaction History
- Fund Transfer

Interface

- Methods Available
 - deposit()
 - withdraw()
 - getBalance()
 - transfer()
 - getTransactions()

Implementation

- HOW operations work
 - Business Logic
 - Validation
 - Database Processing

Benefits of ADT Abstraction

- Modularity
 - Independent modules
- Reusability
 - Same interface in many systems
- Maintainability
 - Easy to update without changing user interface
- Better Software Quality
 - Easy Testing
 - Secure Design
 - Flexible Development
 - Reduced Errors

ADT Separates Logical Design from Implementation

- ADT
 - Defines WHAT to do
 - Hides Internal Implementation
- Implementation
 - Defines HOW it is done
 - Arrays
 - Linked Lists
 - Hash Tables
 - Database Storage

Examples of Banking Operations

- Deposit
 - Add money to account
- Withdrawal
 - Remove money after balance verification
- Balance Inquiry
 - Display current account balance
- Transaction History
 - View previous deposits/withdrawals
- Fund Transfer
 - Move money between accounts

Data Structures

- Arrays
- Linked Lists
- Hash Maps
- Queues
- Trees / Databases